

NullWire Whitepaper v2

Phase 0 Prototype Baseline

NullWire Protocol

April 2026

1 Purpose

NullWire Phase 0 is a prototype of a blockchain-coordinated mixnet messaging system.

This document does not claim that NullWire is already a Tor replacement or that it has solved global traffic analysis. The purpose of Phase 0 is narrower:

- prove that off-chain payload routing through three hops works;
- prove that Solana can coordinate nodes and reward bookkeeping as a control plane;
- measure latency and bandwidth overhead instead of guessing; and
- make the public technical story internally consistent.

2 Architecture Boundary

The architecture now has one hard boundary:

- message payloads stay off-chain;
- Solana stores control-plane state only.

Solana is used for:

- node registry records;
- stake state;
- reward bookkeeping; and
- protocol parameter placeholders.

Solana is not used for:

- message payload persistence;
- user mailboxes;
- sender or receiver identity; or
- communication metadata.

3 Packet Construction

Phase 0 no longer sends a cleartext route list in the wire format.

The current prototype packet model is:

- fixed-size payload body (2048 bytes);
- nested encrypted hop directives;
- one directive peeled per relay or final gateway; and
- next hop revealed only to the current hop.

The sender fetches relay and gateway transport public keys from the registry, then wraps the payload in multiple encrypted layers. Each relay decrypts one layer, learns only the next endpoint, and forwards the remaining opaque packet. The final gateway decrypts the last layer, learns the mailbox name, and stores the fixed-size payload.

This is a real hop-by-hop sealed packet construction. It is still not full Sphinx.

4 Phase 0 Components

Component	Responsibility
Solana control-plane program	Registry state, stake state, reward bookkeeping
Relay service	Decrypt one hop directive, apply delay profile, forward inner packet
Gateway service	Accept packet submission, receive final delivery, store mailbox entries
CLI	Register nodes, fetch directory snapshots, build layered packets, poll mailboxes

5 Registry Record

Each node publishes:

- `node_id`
- `role`
- `endpoint`
- `transport_pubkey_hex`
- `stake_state`
- `last_seen_unix`
- `reputation`

Roles:

- `RelayLayer1`

- RelayLayer2
- RelayLayer3
- Gateway

6 Delay Profiles

The prototype exposes three delay profiles:

- NoDelay
- ReducedPoisson
- LoopixLike

These are test configurations. They are not validated anonymity modes.

7 Test Plan

Functional

- Register three relays and two gateways.
- Fetch a registry snapshot from the CLI.
- Send Alice to Bob through a three-hop route.
- Deliver to Bob's gateway mailbox.
- Poll the mailbox and verify delivery.

Failure

- Stop one relay and confirm forwarding failure is visible to the client.
- Use stale registry data and confirm refresh is required.
- Simulate control-plane unavailability and confirm cached snapshot behavior is explicit.

Measurement

- Measure median and p95 latency for all delay profiles.
- Measure bandwidth overhead with and without cover packets.
- Measure throughput ceiling per relay and per gateway on one host.

8 Explicit Non-Goals

Phase 0 does not include:

- anonymous admission or zk credentials;
- Double Ratchet;
- token launch;
- group messaging;
- file transfer;
- mobile or browser clients; or
- production anonymity claims.

9 Success Criteria

Phase 0 is good enough to continue only if:

- three-hop delivery works repeatedly;
- measured latency is acceptable enough to justify a stronger next phase;
- docs and code describe the same system; and
- external review does not find a fatal architectural flaw.

10 Current Claim

The only clean claim for the project at this stage is:

NullWire is a prototype of a blockchain-coordinated mixnet messaging system.

Anything stronger than that is marketing debt.